VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2011
Higher 1

VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE

CANDIDATE NAME

CLASS

BIOLOGY 8875/01

Paper 1 Multiple Choice

21 September 2011

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.
Do not use any staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided.

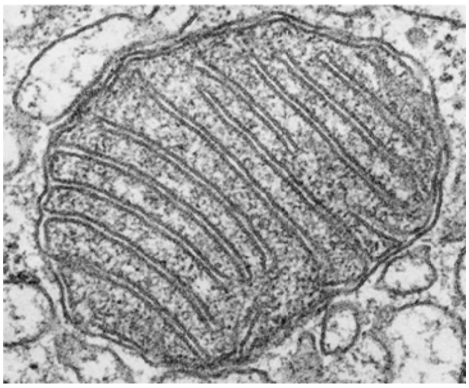
There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deduced for a wrong answer.
Any rough working should be done in this booklet.

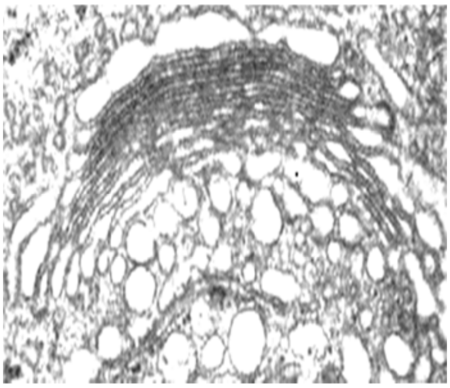
1 Which of the following eukaryotic cell structures contain nucleic acid?

I

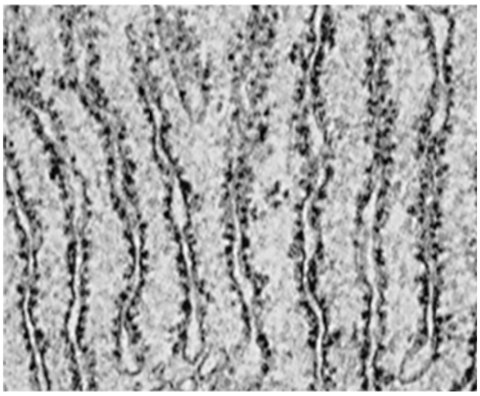


http://antiparos.zoo.ox.ac.uk/presentations/Talks/20060516-Bioimage-DataWeb.html

II

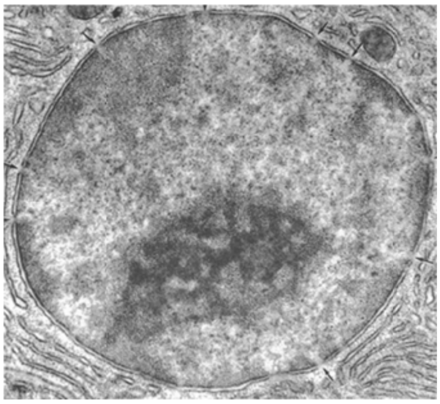


III



http://bass.bio.uci.edu/~hudel/bs99a/lecture22/index.html

IV



http://course1.winona.edu/sberg/241f05/Lec-note/Cells.htm

A IV only

B I and IV

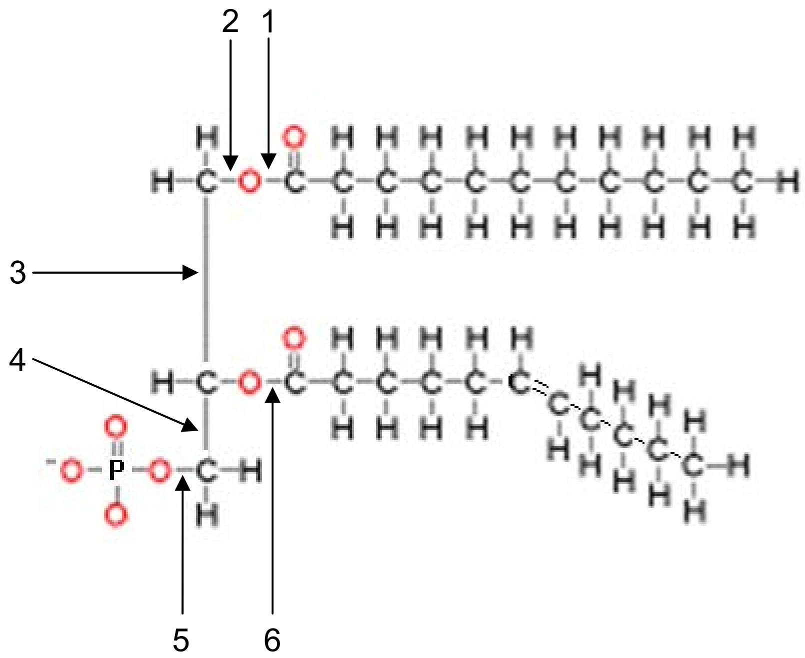
C I, III and IV

D all the above

2 Which of the following options identifies correctly the structures listed below?

	Centriole	**Lysosome**	**Rough endoplasmic reticulum**
A	It is arranged a ring of nine triplets of microtubules attached to each other by fibrils.	It releases its contents for self-digestion of a cell when the cell is damaged.	It is an intracellular double-membrane system to which ribosomes are attached.
B	It is situated next to the nucleus at a region known as centrosome.	It is formed by budding from smooth endoplasmic reticulum.	It originates from the outer membrane of the nuclear envelope.
C	It occurs as a pair at right angles to one another.	It allows increased surface area for metabolism of carbohydrate.	It provides a large surface area for attachment of mRNA.
D	It is composed of microtubules and important for organizing the spindle fibers.	It is formed by budding from Golgi apparatus.	It is a membranous system where proteins synthesized are passed into.

3 The diagram represents a phospholipid.



At which bonds 1 – 6 does hydrolysis occur?

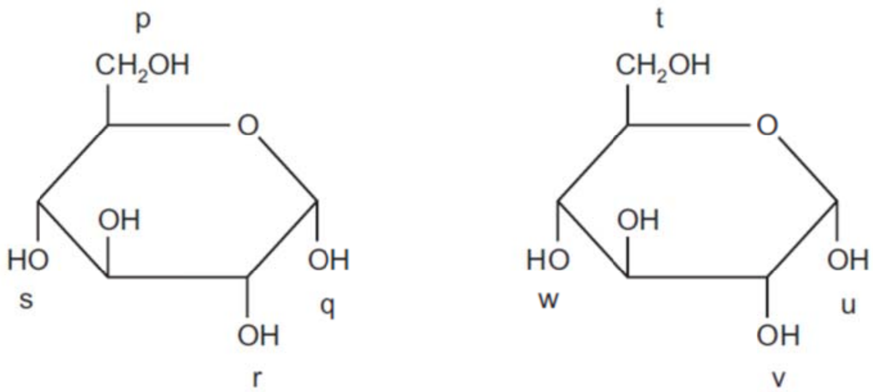
A 1, 3 and 5

B 1, 5 and 6

C 2, 3 and 4

D 2, 4 and 6

4 The diagram shows two molecules of glucose. Four possible bonding positions are labelled p, q, r, and s, and t, u, v, w.



When these two molecules condense to form glycogen, where could bonds form?

A p - u or p - v

B p - u or q - w

C p - v or q - w

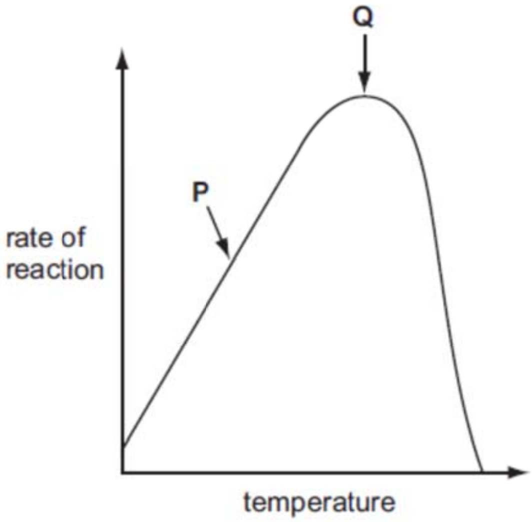
D p - w or v - w

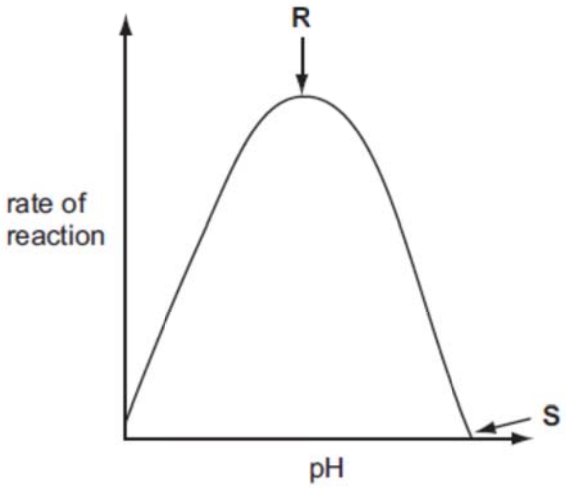
5 Which statement describes how allosteric enzymes work?

	Reversible	Competitive inhibition	End-product inhibition	Active and inactive forms
A	x	x	√	√
B	√	√	x	√
C	√	x	√	√
D	x	√	x	x

Key: √ = yes x = no

6 The graphs show the effects of temperature and pH on enzyme activity.





Which option explains the enzyme activity at the point shown?

	P	Q	R	S
A	The activation energy of the reaction is increasing.	The kinetic energy of enzyme and substrate is highest.	Covalent bonds are formed between enzyme and substrate.	The substrate is completely denatured.
B	Hydrogen bonds are formed between enzyme and substrate.	The kinetic energy of enzyme and substrate is highest.	The number of enzyme-substrate complexes has reached a maximum.	Ionic bonds in the enzyme are broken and active site is lost.
C	Hydrogen bonds are formed between enzyme and substrate.	The number of enzyme-substrate complexes has reached a maximum.	The conformation of the active site is most ideal for binding of substrate.	The enzyme is completely denatured.
D	The activation energy of the reaction is increasing.	Covalent bonds are formed between enzyme and substrate.	The number of denatured enzyme molecules is lowest.	Peptide bonds in the enzyme begin to break.

7 Which of the following do mitosis and the **second** division of meiosis have in common?

1. Chromatids separate during anaphase.
2. Genetically identical material passes to each pole.
3. Centromeres divide at the beginning of anaphase.
4. A ‘resting phase’ (interphase) always occurs before the division commences.
5. Both chromosomes of a homologous pair lie at the equator during metaphase.

A 1 and 3

B 2 and 5

C 1, 2 and 4

D 3, 4 and 5

8 Which of the following statements **does not** apply to DNA replication in bacteria and eukaryotes?

1. Replication begins at a single origin of replication.
2. Replication proceeds in a 5’ to 3’ direction only and is bidirectional from the origin(s).
3. Replication occurs continuously for both stands of new DNA in bacteria since it is a circular DNA.
4. There are numerous different bacterial chromosomes, with replication occurring in each at the same time.

A 2 only

B 2 and 3

C 1 and 4

D 1, 3 and 4

9 Which of the following statements about RNA is **incorrect**?

A Small nuclear RNAs bind with ribonucleoproteins to form spliceosomes.

B Ribosomal RNAs bind with transfer RNA to catalyse the formation of peptide bonds.

C Transfer RNA translates the base sequence present on the DNA into the amino acid sequence in a protein.

D The base sequences in the RNA molecules are complementary to the base sequences in the DNA that is being transcribed.

10 The sequence of DNA bases

GGCAATTGGAAACGAATACCCAGT

codes for the following sequence of amino acids.

glycine- aspargine-trytophan-lysine-arginine-isoleucine-proline-serine

A mutant organism, in which one of the bases was deleted and then inserted at a different point in the DNA molecule, produced a peptide with the following sequence:

glycine- aspargine-trytophan- aspargine -glutamine-isoleucine-proline-serine

Which base was removed by this mutation?

A adenine

B guanine

C cytosine

D thymine

11 In the immune system, B cells are lymphocytes that can produce different antibodies to target foreign agents that invade the body.

Which of the following provides a possible mechanism for the production of the different antibodies in a B cell?

A Activation of different specific transcription factors to regulate gene expression.

B Crossing over and random segregation to create new combination of alleles.

C Splicing of selected alternative exons into final mRNA products.

D Acetylation of histone proteins to create different levels of packaging in chromatin model.

12 Which of the following regarding genes and chromosomes is **incorrect**?

A Their copy numbers in the cell decrease after meiosis, and increase during fertilization

B Both undergo segregation during meiosis

C A gene is a part of the chromosome without the histones.

D Genes can be replicated but chromosomes cannot.

13 Which statement is correct?

1. A phenotype is all the characteristics of an organism that are expressed.
2. A genotype is the genetic constitution of an organism consisting of the alleles present for one or more genes.
3. A phenotype arises from the interaction of an individual's genotype and its environment.
4. A genotype is the sum total of all the genes and their different alleles in a population of organisms.
5. An individual's genotype is inherited from a single parent by asexual reproduction or from two parents via gametes.

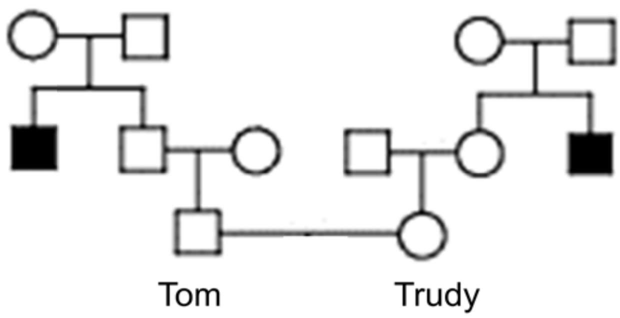
A 1 and 2

B 1 and 3

C 2, 4 and 5

D 1, 2, 3 and 5

14 A married couple Tom and Trudy found out that each had an uncle with Tay-Sachs disease (a rare but severe autosomal recessive disease). The pedigree is as follows:



What is the chance that Tom is a carrier?

A 1 in 2

B 1 in 3

C 1 in 4

D 2 in 3

15 In a particular variety of tomato plant, the allele for red fruit colour is dominant to the allele for yellow fruit colour and the allele for hairy stems is codominant with the allele for hairless stems. Two plants, both heterozygous for both genes are crossed.

How many phenotypes are expected in the next generation?

A 4

B 6

C 8

D 12

16 Which combination correctly describes the dark reactions of photosynthesis?

	site of reaction in chloroplast	ATP	NADP
A	granum	ADP + P_i → ATP	reduced
B	granum	ATP → ADP + P_i	reduced
C	stroma	ADP + P_i → ATP	oxidised
D	stroma	ATP → ADP + P_i	oxidised

17 A metabolic poison rotenone was added to a mixture of homogenized cells with intact chloroplasts. It is discovered that rotenone can bind irreversibly to the electrons carriers on the thylakoid membrane in chloroplasts and alter their conformational shape.

In what way would the Calvin cycle be affected by rotenone poisoning?

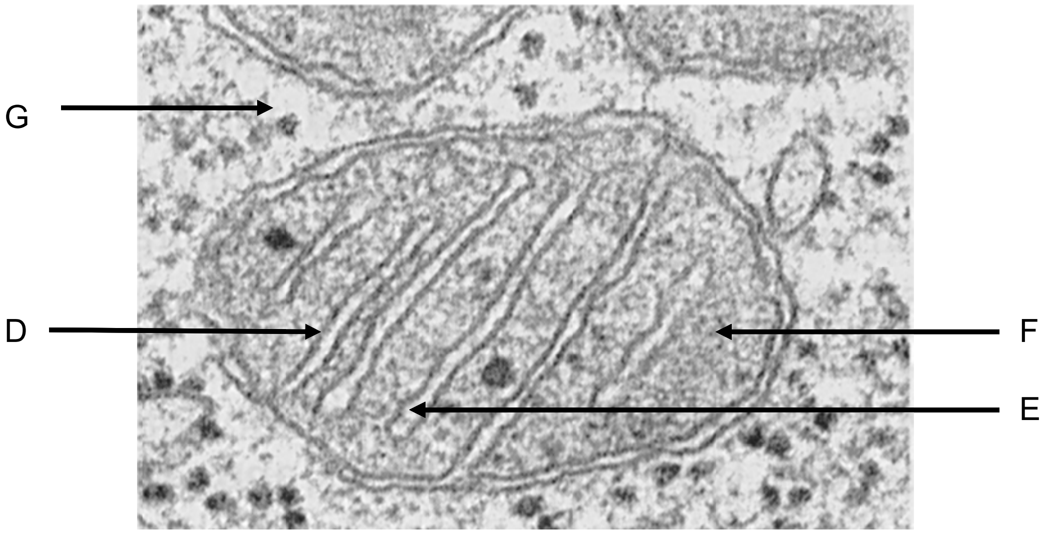
A Calvin cycle will stop, as the accumulation of triose phosphate will create feedback inhibition on glucose formation.

B The rate of Calvin cycle will decrease due to the accumulation of glycerate-3-phosphate.

C The rate of Calvin cycle will increase since electrons can flow directly to final electron acceptor to form reduced NADP.

D There will be no effect since Calvin cycle is an enzyme-controlled process.

18 Match the following processes with structures labeled D – G in the electron micrograph below.



http://www.cbs.dtu.dk/staff/dave/roanoke/bio101ch05.htm

	Drop in pH observed	*Attachment of inorganic phosphate to ADP*	*Series of reduction / oxidation reactions*	*Carbon dioxide evolved*
A	D	E	F	G
B	E	G	D	F
C	G	F	D	E
D	F	G	E	D

19 Which one of the following substances, when added, would directly result in a decline in ATP production in glycolysis?

1. A chemical that would bind to NAD^+ irreversibly and induces its reduction to NADH.
2. An inhibitor that has a similar structure to glucose but cannot be broken down by respiratory enzymes.
3. A chemical that creates an anaerobic environment by combusting in oxygen
4. A reagent that binds to the active site of ATPase permanently

A 1 and 2 only

B 1, 2 and 4 only

C 3 and 4 only

D All of the above

20 Which of the following statements is **false** regarding the polysaccharides starch, glycogen and cellulose?

1. All have 1 → 4 glycosidic linkages.
2. Starch is built from a different monomer compared to the other two polysaccharides.
3. Each is built from a single type of monomer.
4. Starch and glycogen differ from each other only in the extent of their branching.
5. Cellulose serves a structural role as it has high tensile strength.

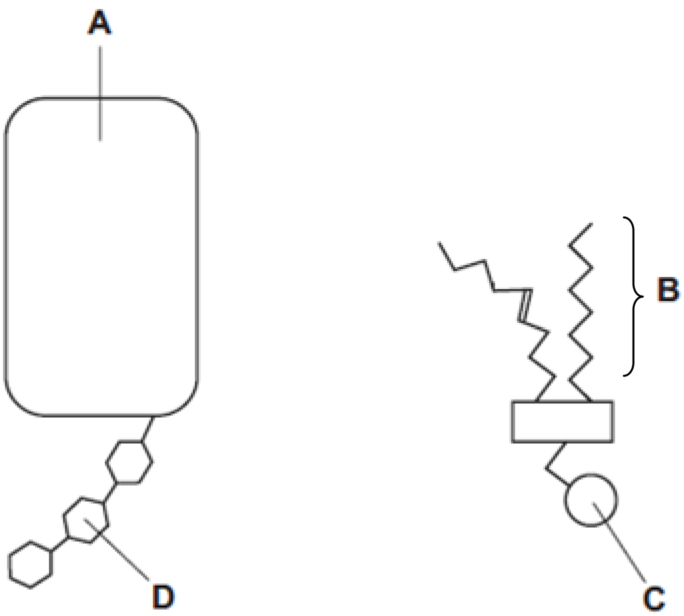
A 1 and 3 only

B 2 and 4 only

C 1, 3 and 5 only

D 3, 4 and 5 only

21 The diagrams show two kinds of molecules found in cell surface membranes.



Which of the following options matches the description shown?

	contains C, H and O	can form hydrogen bond with water	affects membrane fluidity
A	A, B	C, D	A, B
B	A, D	A, C, D	B
C	B, D	A, C	B, D
D	B, C, D	B, D	A

22 The following statements (not in correct order) summarise the steps in natural selection.

1. Some individuals are better suited to a particular environment.
2. Over time there is an increase in particular characteristics in the population.
3. There is variation within a population, some of which is genetic.
4. Individuals better suited to the environment are more successful at survival and reproduction.

The order of statements which best describe natural selection are

A 1, 3, 2, 4

B 1, 2, 4, 3

C 2, 3, 1, 4

D 3, 1, 4, 2

23 *Lucilia cuprina*, the sheep blowfly, lays its eggs in wounds and the wet fleece of sheep. The larvae hatch and burrow into the sheep's skin, causing distress, reduced wool production and sometimes death. Chemicals were used in the past to control the *L. cuprina* but these became less effective as sheep blowfly developed a resistance to the chemicals. The cause of the increased resistance to the chemicals was most likely due to

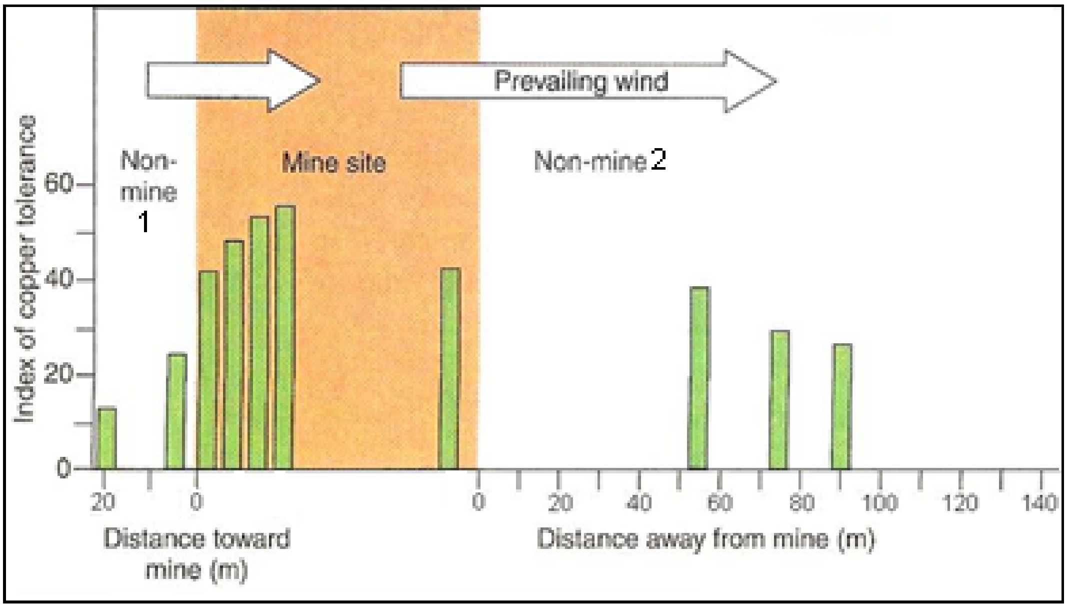
A susceptible sheep dying and leaving behind the more resistant sheep which selects for the more resistant blowfly.

B the insecticide producing a change in a gene which enhanced the survival of the blowfly.

C a chance mutation in a blowfly gene conferring a survival advantage in the chemical environment.

D the insecticide producing a change in phenotype which enhanced reproduction of the blowfly.

24 The figure below shows the degree of copper tolerance in colonial bentgrass growing in old abandoned mine sites in Great Britain. Although mining operations has stopped in these areas, the concentration of copper ions in the soil is still higher compared to other areas. Copper and other heavy metals are generally toxic to plants but alleles of certain genes can confer copper tolerance.



Which of the following could have accounted for the difference in the index of copper tolerance between the bentgrass grown in non-mine sites 1 and 2?
You are to assume that there is no other population of bent grass growing to the left of the non-mine site 1 plants.

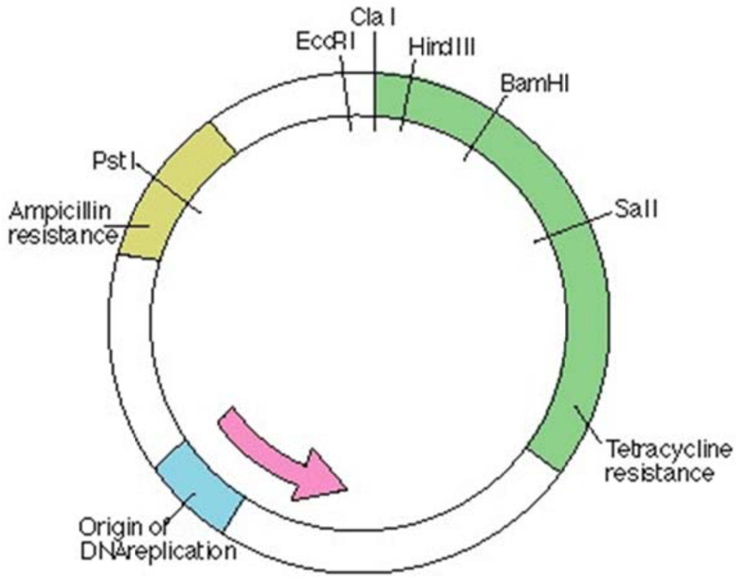
A In site 1, plants having a higher copper tolerance are at a selective disadvantage whereas those growing in site 2 are at a selective advantage.

B In site 1, copper tolerance arises as a result of mutation whereas in site 2 the allele(s) is(are) introduced into the population via pollen carrying the allele from the mine site that have undergone natural selection.

C In site 2, genetic recombination has occurred resulting in a higher level of copper tolerance in the population by the pollen introduced from the mine site.

D In site 2, copper tolerance is a result of polygenes whereas in site 1, it is due to one gene with 2 alleles.

25 The following diagram shows a plasmid which is used for genetic engineering purposes.



A eukaryotic gene is inserted in the plasmid.

Which of the following is correct of bacteria cells transformed with this recombinant plasmid?

	Restriction enzyme used	Condition which the transformed bacteria cells with recombinant plasmid is grown in
A	*Pst*I	Medium containing ampicillin only.
B	*Sal*I	Medium containing tetracycline only.
C	*BamH*I	Medium containing ampicillin only.
D	*Hind*III	Medium containing tetracycline only.

26 Which of the following shows a correct comparison of the two processes?

	Polymerase chain reaction	**Translation**
A	Occurs in the nucleus	Occurs in the cytoplasm
B	Requires ribonucleic acids	Does not require deoxyribonucleic acids
C	Synthesizes selected section of the DNA	Synthesizes selected section of the mRNA
D	Uses information on the template between the primers	Uses information on the template between the start and stop codons

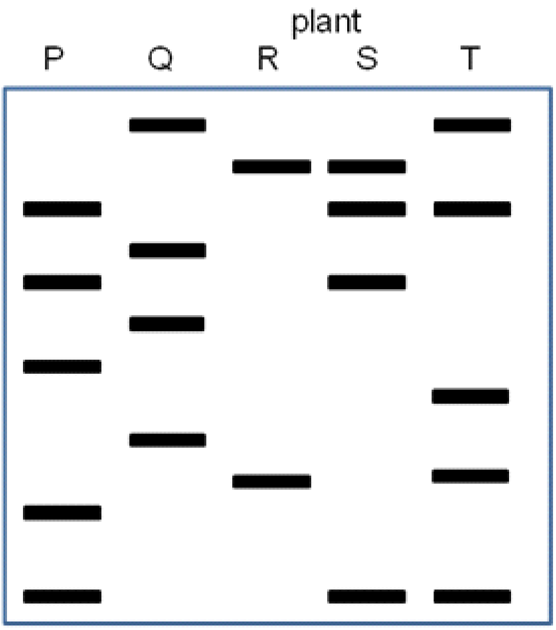
27 A possible route of escape of an inserted gene from a genetically engineered crop of oil-seed rape, *Brassica napus*, is into populations of the wild turnip, *Brassica rapa*.

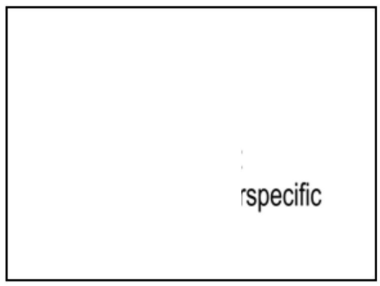
Two populations of wild turnip growing next to large fields of a cultivar of oil-seed rape were studied. Seeds were collected from these wild turnips plants and their DNA analysed using restriction enzymes and electrophoresis.

Using the key provided below, determine

(i) whether plants S and/or T are interspecific hybrids and

(ii) the parental plants they are derived from





	Interspecific hybrid	*Parent plants*
A	Both S and T	S: P x R T: P x Q
B	Both S and T	S: P x Q T: P x R
C	S only	P x R
D	T only	Q x R

28 Some of the goals of the Human Genome Project are listed.

1. To determine the base sequence of the whole of the human genome.
2. To identify the genes encoded by the base sequence.
3. To find the locations of the genes on the 23 pairs of human chromosome.

Which goals had to be achieved to allow the development of a single-stranded DNA probe for detecting the presence of a mutant allele on human chromosome 7?

A 1 and 2

B 1 and 3

C 2 and 3

D 1, 2 and 3

29 Which of the following regarding embryonic stem cells and hematopoietic stem cell is true?

A As embryonic stem cells develop, they turned into hematopoietic stem cells as they lose their ability to differentiate into all types of cells.

B Embryonic stem cells have more genes than hematopoietic stem cells and thus are able to form more types of cells.

C Under normal conditions, embryonic stem cells express more of their genes compared to the hematopoietic stem cells.

D Both stem cells are derived from the zygotic stem cells with the hematopoietic stem cells having a lowered telomerase activity compared to the embryonic stem cells.

30 Which of the following is **not** a concern with growing genetically engineered crops e.g. *Bt* corn?

A Toxic effects of *Bt* on non-target lepidopteran insects e.g. monarch butterfly larvae (of the insect order *Lepidoptera*) could upset ecological balance.

B Transfer of selective marker to bacteria that reside in the gut of humans thereby rendering the bacteria antibiotic resistant.

C Flow of the *Bt* gene from cultivated corn to wild relative which could then lead to loss of biodiversity.

D Transfer of *Bt* gene to the pests thereby increasing their resistance to the *Bt* toxin